MIL-D-10396D
15 September 1982
SUPERSEDING
MIL-D-10396C
2 December 1969

MILITARY SPECIFICATION

DIMETHYL HYDROGEN PHOSPHITE, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two grades of dimethyl hydrogen phosphite, $(CH_3O)_2P(OH)$.

1.2 Classification. Dimethyl hydrogen phosphite shall be of the following grades as specified (see 6.2):

Grade A - see table I
Grade B - see table I

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1168 - Lot Numbering of Ammunition

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D941 - Density and Relative Density (Specific Gravity) of Liquids by Lipkin Bicapillary Pycnometer
D1193 - Reagent Water
D1209 - Color of Clear Liquids (Platinum-Cobalt Scale)
D1218 - Refractive Index and Refractive Dispersion of Hydrocarbon Liquids

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(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Chemical and physical characteristics. The dimethyl hydrogen phosphite shall conform to the applicable chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Grade A	Grade B	Test paragraph
Appearance	Clear and free from suspended matter	Clear and free from suspended matter	4.2.4.1
Color No., platinum-cobalt scale	5	50	4.2.4.2
Specific gravity, 25°/4° C	1.190 ± 0.005	1.190 ± 0.005	4.2.4.3
Refractive index (N_D 25° C)	1.400 to 1.403	1.400 to 1.403	4.2.4.4
Chlorine, percent by weight, max	0.05	0.05	4.2.4.5
Trivalent phosphorus (before acid hydrolysis), percent by weight, min	27.59	26.74	4.2.4.6(a)
Trivalent phosphorus in monomethyl hydrogen phosphite, percent by weight, max	0.08	0.32	4.2.4.6(c)
Methoxy, percent be weight, min	55.3	53.9	4.2.4.7

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the dimethyl hydrogen phosphite produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test (see 6.5). Sampling shall be conducted in accordance with table II. A representative specimen of approximately 1 quart shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for tests

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing or illegible

4.2.3.2 For test (see 6.5). Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 1 quart. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

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4.2.4 Tests (see 6.5). Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Appearance. Visually examine the specimen for clarity and suspended matter.

4.2.4.2 Color. Determine the color of the specimen in accordance with ASTM D1209.

4.2.4.3 Specific gravity. Determine the specific gravity of the specimen in accordance with ASTM D941, at a test temperature of 25°C and a reference temperature of 4°C.

4.2.4.4 Refractive index. Determine the refractive index of the specimen in accordance with ASTM D1218 at a temperature of 25°C.

4.2.4.5 Chlorine. Weigh to the nearest milligram approximately 0.5 g of the specimen in a glass sampling bulb. Place the bulb containing the specimen in a heavy walled, glass-stoppered 500-ml Erlenmeyer flask containing 50 ml of 10-percent sodium hydroxide solution. Break the bulb by vigorously shaking the flask and continue shaking until all vapors in the flask are absorbed by the sodium hydroxide solution. Remove the stopper from the flask and by use of a glass rod crush the stem of the bulb. Replace the stopper and allow to stand approximately 30 minutes. Acidify with nitric acid. From a buret add 0.1N silver nitrate solution in definite excess. Filter and wash the precipitate with cold water until free of silver nitrate. To the filtrate and washings add 2 ml of a saturated solution of ferric ammonium sulfate or a 10-percent solution of ferric sulfate. Titrate the solution with 0.1N potassium thiocyanate solution to a permanent reddish-brown end point. Calculate the percent chlorine as follows:

$$\text{Percent chlorine} = \frac{3.546\ (AB - CD)}{W}$$

where: A = Milliliters of silver nitrate solution used,
B = Normality of silver nitrate solution,
C = Milliliters of potassium thiocyanate solution,
D = Normality of potassium thiocyanate solution, and
W = Weight in grams of specimen.

4.2.4.6 Trivalent phosphorus.

(a) Before acid hydrolysis. To a 500-ml iodine titration flask add 50 ml of freshly prepared saturated sodium bicarbonate solution, 50.0 ml of 0.1N iodine solution measured from a buret, and bulbed sample of the specimen weighing 0.1000 to 0.1500 g. Break the bulb starting at the capillary end with a glass rod flattened at one end. Wash down rod with several milliliters of

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water. Swirl, and allow to stand in the dark for 30 minutes using a water seal in the flask well around the stopper. Titrate the excess iodine with 0.1N sodium arsenite solution to a colorless end point with or without starch solution as an indicator. Calculate the percent trivalent phosphorus before acid hydrolysis as follows:

$$\text{Percent trivalent phosphorus} = \frac{1.549\ A\ (B - C)}{W}$$

where: A = Normality of sodium arsenite solution,
B = Milliliters of sodium arsenite solution used to titrate blank,
C = Milliliters of sodium arsenite solution used to titrate specimen, and
W = Weight in grams of specimen.

(b) After acid hydrolysis. Bulb and weigh a 0.1000 to 0.1500 g sample of the specimen. Place the bulb into a 300-ml Erlenmeyer flask with a 24/40 tapered ground glass opening containing 15 ml of 1N hydrochloric acid. Break the bulb starting at capillary end with a glass rod flattened at one end, and wash down the rod with 3 to 5 ml of water. Reflux gently under a water-cooled condenser for 1 hour. Cool, wash down the condenser with 3 to 5 ml of water, and add approximately 7 g of powdered sodium bicarbonate. From a buret add 50.0 ml of 0.1N iodine solution and swirl. Allow to stand in the dark for 30 minutes using a water seal in the flask well around the stopper. Cautiously remove the stopper to avoid loss of iodine and back titrate with 0.1N sodum arsenite solution to a colorless end point with or without starch solution as an indicator. Calculate the percent trivalent phosphorus after acid hydrolysis as follows:

$$\text{Percent trivalent phosphorus} = \frac{1.549\ A\ (B - C)}{W}$$

where: A = Normality of sodium arsenite solution;
B = Milliliters of sodium arsenite solution used to titrate blank,
C = Milliliters of sodium arsenite solution used to titrate specimen, and
W = Weight in grams of specimen.

(c) In monomethyl hydrogen phosphite. Calculate the percent trivalent phosphorus due to monomethyl hydrogen phosphite as follows:

$$\text{Percent trivalent phosphorus} = A - B$$

where: A = Percent trivalent phosphorus after acid hydrolysis calculated in 4.2.4.6(b), and
B = Percent trivalent phosphorus before acid hydrolysis calculated in 4.2.4.6(a).

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4.2.4.7 Methoxy determination.

(a) Apparatus. Assemble the apparatus as shown on figure 1.

(b) Procedure. Pipet a sufficient volume of wash solution, consisting of 10 percent finely divided red phosphorus suspended in 15-percent cadmium sulfate solution in water, to a height level with the bottom of the side outlet tube on the absorber side of the bubble counter. Insert the stopper in the bubble counter. Add 20 to 25 ml of a 10-percent solution of anhydrous sodium acetate in glacial acetic acid and 1 ml of bromine to the absorbing column. Assemble the apparatus except for the reaction flask. Place a small, thin-walled, sealed glass bulb containing 0.0500 to 0.0800 g of specimen into the reaction flask. From a 10-ml pipet add 5 ml of hydriodic acid (special for micro-analysis, Pregl), while breaking the bulb with a glass rod, the bottom of which has been flattened, crushing the stem of the bulb first, then the rest of the bulb. Wash the rod down with the remaining 5 ml of the hydriodic acid. Care is required to insure that the specimen is covered with hydriodic acid at all times while breaking the bulb. Assemble the reaction flask to the apparatus. Connect the carbon dioxide tube to the reaction flask, and adjust the carbon dioxide rate so that one to two bubbles per second pass through the bubble counter. Seal the glass joints of the reaction flask with hydriodic acid. Use an electrically heated oil bath, and raise the temperature of the bath to 135° to 140° C over the period of 1 hour. Continue to heat at that temperature for approximately 1 hour (total time 2 hours). Wash the contents of the absorber with water into a 500-ml iodine flask containing 10 ml of the above anhydrous sodium acetate solution in glacial acetic acid. The total volume should not exceed 150 ml. Swirl and let stand 10 minutes. Decolorize the free bromine in the iodine flask by the dropwise addition of formic acid, swirling the flask during the addition. Stopper the flask and shake to remove bromine vapor. If the solution does not decolorize to a water-white color, discard the solution and repeat the analysis, as accurate results will not be obtained. When the solution is properly decolorized, add approximately 2 g of solid potassium iodide. Shake the flask until all of the potassium iodide is in solution. Add water around the glass stopper to form a water seal. Allow to stand in the dark for 10 minutes. After standing, remove from the dark and cool the flask to below room temperature to prevent loss when removing the stopper. Remove the stopper, wash down with water, and titrate the liberated iodine with 0.1N sodium thiosulfate solution to a light yellow color, swirling continuously. Calculate the percent methoxy as follows:

$$\text{Percent methoxy} = \frac{0.5172\ A\ B}{W}$$

where: A = Milliliters of sodium thiosulfate solution used,
B = Normality of sodium thiosulfate solution, and
W = Weight in grams of specimen.

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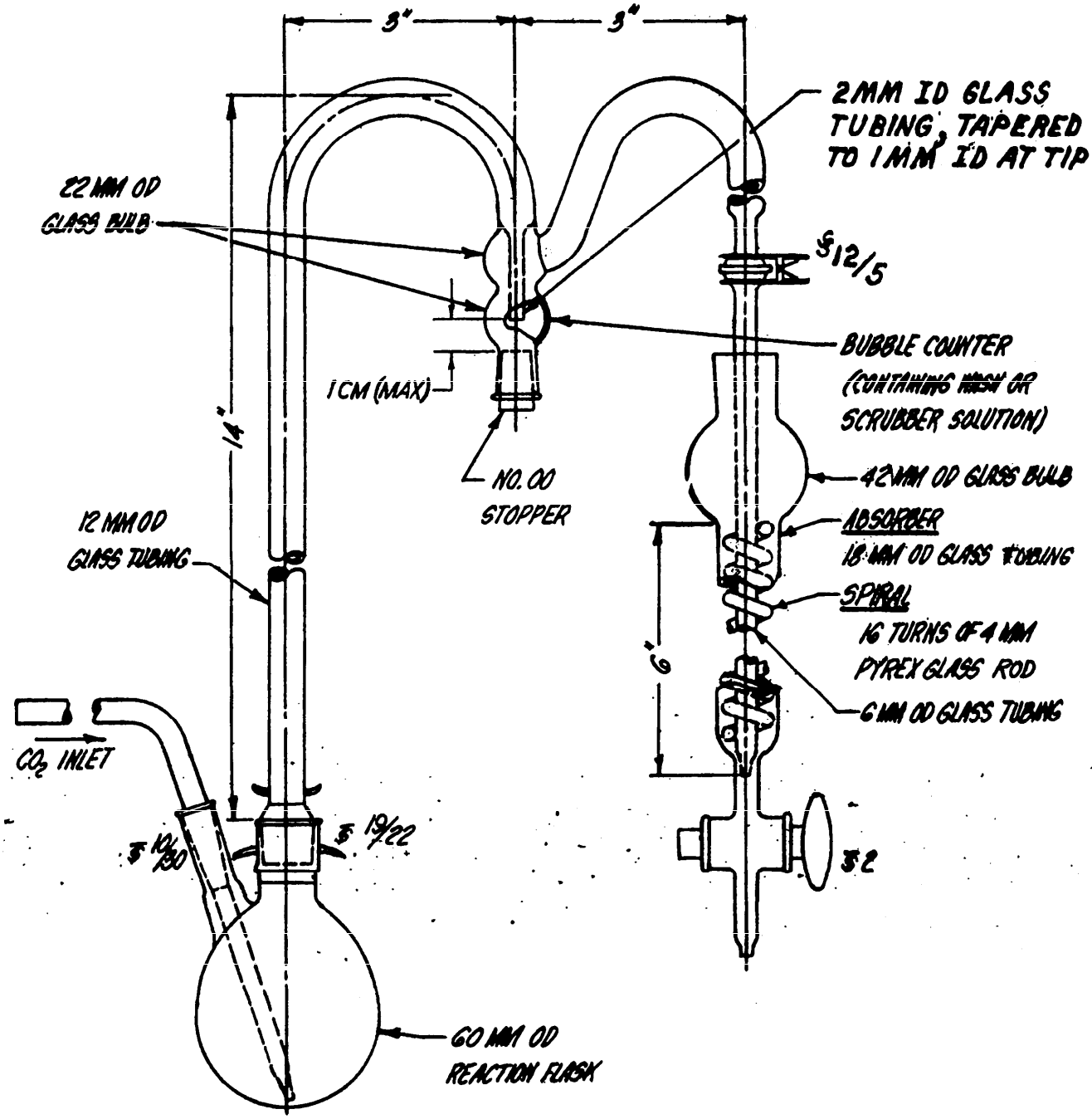


Figure 1 Methoxy apparatus

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5. PACKAGING

5.1 Unit packing, level C. The dimethyl hydrogen phosphite shall be unit packed level C in accordance with Department of Transportation (DOT) regulations in a 1 or 54 gallon (gal) quantity, as specified (see 6.2).

5.1.1 One-gal quantity. A quantity of 1 gal (+1-1/3 or -0) fluid ounces of dimethyl hydrogen phosphite shall be unit packed in a nominal 1-gal capacity glass bottle conforming to group A, class 1, type d, grade 2 of PPP-C-186 and furnished with closure A screw-cap and outer seal A. The bottle shall be closed so as to be leak-proof in accordance with the bottle manufacturer's instructions.

5.1.2 Fifty-four gal quantity. A quantity of 54 (+1/2 or -0) gal of dimethyl hydrogen phosphite shall be unit packed level C in a clean dry nominal 55-gal capacity polyethylene drum within a steel drum conforming to DOT specifications 2SL and 37M, respectively. The drum components shall be closed in accordance with the manufacturer's instructions so as to be free of leakage.

5.2 Packing, level C.

5.2.1 One-gal quantity. Four 1-gal bottles of dimethyl hydrogen phosphite shall be packed in a two bottle by two bottle layer, with closure end uppermost, within a box. The box shall meet the minimum requirements of DOT specification 12A. The entire pack shall comply with all applicable regulatory requirements and shall be in accordance with Uniform Freight Classification Rules, National Motor Freight Classification Rules, and the rules applicable to any other selected mode of transportation.

5.2.2 Fifty-four gal quantity. The 54-gal quantity, unit packed as specified in 5.1.2, shall require no further protection for shipment.

5.3 Unitization. The 1-gal packs of dimethyl hydrogen phosphite shall be palletized in accordance with the type I load of MIL-STD-147.

5.4 Marking. Containers and pallet loads of dimethyl hydrogen phosphite shall be marked in accordance with MIL-STD-129, DOT regulations, and any other applicable regulations. In addition, each unit pack and shipping container shall be labeled to show the following precautionary information:

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HAZARD:

Irritant, Eyes and Skin

WARNING! CAUSES IRRITATION
Do not breathe vapors.
Use with adequate ventilation, gloves, and face shield.
Avoid contact with skin and eyes. Wash thoroughly after handling.

FIRST AID: In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. (Wash clothing before reuse.) For eyes get medical attention.

6. NOTES

6.1 Intended use. Dimethyl hydrogen phosphite is intended for use as an intermediate in chemical agent production and also as an agent simulant.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Grade required (see 1.2)
(c) Unit quantity required (see 5.1)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

6.5 Sampling and testing precautions. This specification covers inspection and use of chemical material which is potentially hazardous to personnel. Dimethyl hydrogen phosphite is hazardous to handle, an irritant to eyes and skin. Personnel sampling or testing this and other highly caustic materials should be adequately protected against contact with the liquid and inhalation of the vapors. Eye protection should be used and eye lavages should be available in laboratory areas. All applicable safety rules, regulations, and procedures must be followed in the handling and processing of this material. Sampling and testing should be performed as rapidly as possible because the material decomposes in contact with atmospheric moisture.

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